THE SOLUTION BOOK

MY AUTOMATED AI SYSTEM

One worked build — follow along.

One fully worked example, so you can see what a finished Week 4 system looks like — an agent brain, a Claude Routine, an N8N workflow, connected tools, and a voice layer.

YOUR SYSTEM HAS FIVE PARTS

N8N / Make workflow | Claude Routines | MCP × 3+ tools | Voice layer | Runs without you

WHAT, WHY, WHERE, WHEN & HOW

The whole assignment in five plain-English lines.

WHAT	Build a complete **Automated AI System** — a multi-step N8N or Make workflow with Claude as the intelligence layer, Claude Routines for the native steps, MCP connections to 3+ tools, and a voice layer (VAPI or ElevenLabs). It solves one real problem and runs end to end without you.
WHY	This is the leap from producing outputs to building infrastructure — agents that reason, decide and act on their own. It unlocks the AI Systems Operator identity and everything you build after.
WHERE	Claude (Projects · Routines · Schedules · MCPs · API) is the brain; N8N / Make is the wiring; VAPI / ElevenLabs is the voice; Notion, Airtable and Slack are where the data lives.
WHEN	Week 4 of the Fellowship — about 90 minutes. Build it for a real use case: student onboarding, customer intake, a content pipeline, a research assistant, or an appointment system.
HOW	Design the agent (goal · tools · memory · escalation) → build a 5-step Claude Routine → wire it into N8N/Make with the Claude API returning JSON → connect 3+ tools by MCP → add a voice layer → trigger it and let it run.

YOUR TOOLBOX – CLAUDE

The Claude power-tools that make the brain of your system.

R Routines

SIDEBAR → ROUTINES

Chain several prompts into one automated sequence: research → summarise → format → store → notify.

S Schedules

SIDEBAR → SCHEDULED

Run a Routine on a cadence — set once, it runs forever (daily brief, weekly digest).

M MCPs & Connectors

CUSTOMIZE → CONNECTORS

Give Claude real-time access to Notion, Slack, Gmail and Calendar inside a Routine.

A Claude API

THE BRAIN NODE

Claude called as the intelligence node inside N8N/Make — returns strict, structured JSON.

YOUR TOOLBOX — WIRING & VOICE

The platforms that carry your system's work, voice and data.

W N8N / Make

THE WIRING

Visual workflow: triggers → actions → conditions. The HTTP node calls Claude; IF/Switch routes the result.

V VAPI · voice agents

THE VOICE

Phone numbers, webhooks, Claude integration, call analytics (Bland AI, Retell AI too).

E ElevenLabs

VOICE OUTPUT

Voice generation and cloning (with consent) — the spoken layer of your system.

D Airtable · Notion · Slack

THE DATA

A structured store, a records/media hub, and notifications — where your system's data lives.

CONNECT & AUTOMATE (EXAMPLES)

1 Connect an MCP tool — Notion & Slack

1. Customize → Connectors (or "+" → Connectors → Manage).
2. Pick Notion (and Slack) → Connect → approve in the app's own window (OAuth).
3. Grant the least access it needs; toggle it on in the chat / Routine.
4. Now Claude can log to Notion and post to Slack inside a Routine.

Safety — Claude never sees your password, and it never sends or changes anything until you say yes.

2 Make a Routine + Schedule — the 5-step agent

1. Sidebar → Routines → New (Create with Claude or set up manually).
2. Chain 5 steps: research → summarise → format as JSON → store to Notion → notify Slack.
3. Sidebar → Scheduled → attach the Routine → pick a cadence + approval.
4. Save — it runs in the cloud on its trigger and reports back.

Like this — a Routine is your agent's body; the Schedule is its heartbeat.

CREATE TWO SKILLS (EXAMPLES)

A Skill is a saved recipe Claude reloads every time — so your agent logic stays consistent.

/ /agent-brain

THE BRAIN

1. Customize → Skills — or in Cowork: "+" → Record a skill.
2. Write a SKILL.md: name, a one-line description (the trigger), and the agent's Goal · Tools · Memory · Escalation as the rules.
3. Zip the folder (folder as root) → Upload → enable.
4. Load /agent-brain at the top of every Routine so the logic never drifts.

/ /voice-script

THE VOICE

1. Create a second Skill that writes voice scripts with prosody markers and pause notation.
2. Its rules: short spoken turns, confirmations, a graceful fallback line.
3. Run /voice-script to generate the lines your VAPI/ElevenLabs agent speaks.
4. Edit and re-upload any time; update the description if the trigger changes.

DESIGN THE AGENT

Every agent runs the same loop — and every agent needs the same four instructions.

THE AGENT LOOP

Perception → Reasoning → Action → Memory ↻

THE AGENT SYSTEM-PROMPT RECIPE

GOAL

Answer course FAQs by voice and log every question asked.

TOOLS

Notion (log questions), Gmail (weekly summary), a course knowledge base.

MEMORY

A Notion database of past questions (external memory) it reads before answering.

ESCALATION

If confidence is low, say so and flag a human — never guess.

Why it works — Goal keeps it on task, Tools bound what it can do, Memory stops it repeating, and Escalation keeps a human in the loop when it's unsure.

WRITE THE BRAIN IN STAR

STAR = Situation · Task · Action · Result. The Claude step inside your workflow is a prompt too — write it in STAR so it returns clean JSON every time. Here is one worked example.

S · Situation

My N8N workflow sends Claude a new lead's call transcript.

T · Task

Qualify the lead and decide where it should go.

A · Action

Score fit 1–5, extract name / need / urgency, choose a route; return STRICT JSON only, no prose.

R · Result

```
{"score":4,"name":"…","need":"…","route":"sales|nurture|drop"}
```

Why it works — the Action pins the exact JSON shape, so N8N's IF/Switch node can route on it without ever breaking.

THE BUILD — PICK ONE

Milestone Project 3 — a live, end-to-end system that operates without you.

- **A · AI Student Assistant** — A voice agent that answers course FAQs, logs questions to Notion, and emails weekly summaries.
- **B · AI Startup Intake System** — A voice agent collects leads, qualifies them via Claude, stores to Airtable, sends personalised follow-ups.
- **C · AI Content Pipeline** — Researches topics, generates scripts, produces audio via ElevenLabs, and publishes to a Notion dashboard.
- **D · Your own design** — Must include Claude Routines, a voice layer, 3+ connected tools, and autonomous operation.

Must include

Claude Routines · a voice layer (VAPI or ElevenLabs) · MCP connections to 3+ tools · autonomous operation on a trigger. Tools: Claude (Projects, Routines, Schedules, MCPs, API), VAPI/ElevenLabs, N8N/Make, Notion, Airtable, plus tools for your use case.

Submit as — a live system + a demo video that shows it running, a voice interaction, and the data output.

NOW BUILD IT IN CLAUDE

1. Pick one use case — A student assistant, B intake system, C content pipeline, or D your own.
2. Design the agent: write its system prompt as Goal · Tools · Memory · Escalation, and save it as /agent-brain.
3. Build a 5-step Claude Routine: research → summarise → format as JSON → store to Notion → notify Slack.
4. Wire it into N8N or Make: a webhook trigger → the Claude API (HTTP node) returns JSON → conditional routing → Notion + Airtable.
5. Add the voice layer with VAPI or ElevenLabs — a live call, or a generated voice response.
6. Put it on a Schedule or trigger, run it end to end, and record a short demo video.

WHAT GOOD LOOKS LIKE

- Runs autonomously and reliably on its trigger
- Claude returns clean, structured JSON every time
- A real use case — someone would actually use it

COMMON MISSES

- Needs you to nudge it through each step (not autonomous)
- Voice or data breaks between tools
- A demo with no real use case behind it

That's a full Week 4 system. Yours can differ — grade the thinking, not the match. menler